

The pline Package, Version 0.01

<https://github.com/Pseudonym321/TikZ-Animations/tree/master1/TikZ/pline>

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Chapter 1

Introduction

pline is a Lua-based 3D graphics package for TikZ.

1.1 What are the current capabilities of pline?

pline can render multiple non-intersecting triangulated mesh surfaces at once. It is also possible to draw a plane which has been clipped by a rectangular prism. There are commands for drawing and labelling geometric vectors too. Curves and surfaces can be defined using matrix transformations.

1.2 How does pline improve on existing packages?

There are two significant packages for 3D artwork in TikZ: pgfplots and tikz-3dplot. pgfplots can z-sort the faces of a single parametric surface, but not multiple surfaces at once. Both pgfplots and tikz-3dplot are capable of changing the camera view, but the result is necessarily orthographic, and the rotations are restricted to azimuth and elevation; that is, the z -axis is always vertical. tikz-3dplot offers useful pgfmath capabilities for doing vector operations. Neither pgfplots nor tikz-3dplot enable matrix transformations. pline is built to render multiple parametric objects at once, enable arbitrary object orientations in non-orthographic projections, and enable matrix transformations.

1.3 Where is the package at, and where it is heading?

The vision of the package pline is to one day be able to render arbitrary numbers of intersecting polygons, by clipping them with each other; this would enable arbitrary surface intersections, without visual artefacts. A more near term goal of pline is to also implement an ODE solving tool which approximates trajectories in vector fields. I also want to add capabilities for doing programming in Lua, instead of in pgf, essentially replacing the pgfmath capabilities of tikz-3dplot with faster Lua-based capabilities. Soon there will also be proper camera culling for projective scenes; that is, the action of not drawing segments which are behind the viewing plane.

Chapter 2

Geometric Vectors

2.1 How are geometric vectors defined in pline?

Geometric vectors are defined by two things: a start-coordinate, and a relative displacement. There are two classifications of geometric vectors; there are zero vectors, and non-zero vectors. A zero vector is represented by a small circle. A non-zero vector is represented by a line segment with an arrow. The endpoints of a non-zero vector may or may not have a circle centered on the endpoint. If there is no circle drawn, the arrow stops directly at the endpoint. If a circle is drawn, then the arrow tip is moved back by the circle's radius. Observe the tips of the vectors in Figure 2.1.1. Both tips lie on the same point, despite one arrow tip being offset by the circle's radius. The syntax for drawing Figure 2.1.1 is shown in the following code.

```
\begin{tikzpicture}
  \drawvector[points = both,yellow]{0,0}{2,0}
  \drawvector[blue]{4,0}{-2,0}
\end{tikzpicture}
```

There are four values for the key points: neither, behind, front and both. These tell the command `\drawvector` whether and where to put circles over the endpoints. See Figure 2.1.2, which uses the following code.

```
\begin{tikzpicture}
  \drawvector[points = neither]{0,0}{0,1} % a)
  \drawvector[points = behind]{1,0}{0,1} % b)
  \drawvector[points = front]{2,0}{0,1} % c)
  \drawvector[points = both]{3,0}{0,1} % d)
\end{tikzpicture}
```

The `\drawvector` syntax is mostly meant for diagrams which don't use z-sorting. Still, they can be used in some 3D still-diagrams; for instance, these vectors connect nicely. They will also eventually

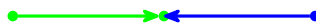


Figure 2.1.1:
The exact location of a geometric vector's tip.

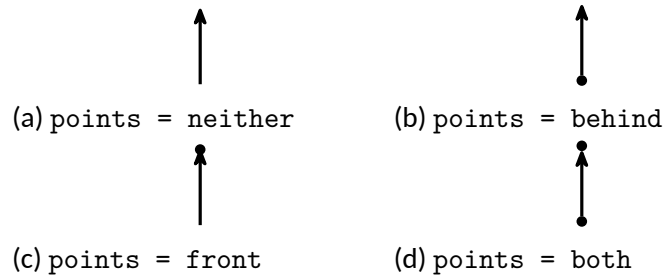


Figure 2.1.2:
The values for the key points.



Figure 2.1.3:
Zero-vector

be able to be drawn flat on planes. There is also a zero-vector command available via `\drawpoint`. See Figure 2.1.3, with the following code.

```
\begin{tikzpicture}
  \drawpoint[red]{0,0}
\end{tikzpicture}
```

Chapter 3

Parametric Objects

3.1 How is a parametric object defined?

Parametric objects are defined using a handful of different pieces of information. In particular, they use parametric functions, domain parameters and a samples parameter for each dimension in the domain. Of course, there are also parameters for color and such as well. Figure 3.1.1 is defined by the following code:

```
\begin{tikzpicture}
  \appendcurve[
    u min = 0
    ,u max = {\tau}
    ,u samples = 200
    ,transformation = {euler(pi/2,pi/3,pi/2)}
    ,x = {2*cos(u)}
    ,y = {2*sin(u)}
    ,z = {u}
    ,draw options = {
      purple
      ,ultra thick
      ,line join = round
    }
  ]
  \appendsurface[
    u min = 0
    ,u max = {\tau}
    ,u samples = 36
    ,v min = 0
    ,v max = {\tau}
    ,v samples = 36*2/3
    ,transformation = {euler(pi/2,pi/3,pi/2)}
    ,x = {2*cos(u)+sphere(u,v)[1][1]}
    ,y = {2*sin(u)+sphere(u,v)[1][2]}
```

```

,z = {u+sphere(u,v)[1][3]}
,draw options = {
    draw
    ,blue,ultra thin
    ,line join = round
}
,fill options = {
    fill = green
    ,fill opacity = 0.7
}
]
\foreach \u in {0,...,35} {
    \foreach \v in {0,...,35} {
        \pgfmathsetmacro{\uscale}{2*pi/(36-1)}
        \pgfmathsetmacro{\vscale}{2*pi/(36*2/3-1)}
        \pgfmathsetmacro{\u}{\u*\uscale}
        \pgfmathsetmacro{\v}{\v*\vscale}
        \appendcurve[
            u min = 0
            ,u max = 0.3
            ,u samples = 2
            ,transformation = {euler(pi/2,pi/3,pi/2)}
            ,x = {
                2 *
                cos(token.get_macro("u")) +
                (u + 1) *
                sphere(
                    token.get_macro("u")
                    ,token.get_macro("v")
                )[1][1]
            }
            ,y = {
                2 *
                sin(token.get_macro("u")) +
                (u + 1) *
                sphere(
                    token.get_macro("u")
                    ,token.get_macro("v")
                )[1][2]
            }
            ,z = {
                token.get_macro("u") +
                (u + 1) *
                sphere(
                    token.get_macro("u")

```

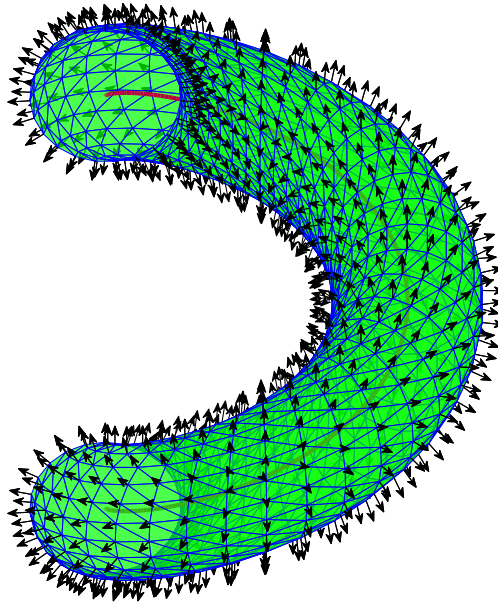


Figure 3.1.1:
A triangulated mesh surface and a curve.

```

,token.get_macro("v")
) [1] [3]
}
,draw options = {-{Stealth[round]}}
]
}
}
\rendersegments
\end{tikzpicture}

```

pline is also capable of handling division by zero without erroring. This is because the math is handled in Lua, where division by zero doesn't error. We clip our results to be well within the scope of the TizZ canvas. The code for Figure 3.1.2 is as follows.

```

\begin{tikzpicture}
\clip (-5,-5) rectangle (5,5);
\pgfmathsetmacro{\angle}{0}
\let\angle\angle
\foreach \longitude in {1,...,36}{
\let\longitude\longitude
\appendcurve[
u min = 0
,u max = {pi}
,u samples = 4*45
,transformation = {euler(pi/2,pi/3,pi/6)}
,x = {

```



```

stereographic_projection(
    matrix_multiply(
        sphere(
            tau *
            token.get_macro("longitude") /
            36
            ,u
        )
        ,euler(
            0
            ,pi/2
            ,(2*pi/24) *
            token.get_macro("angle") /
            36
        )
    )
) [1] [1]
}
,y = {
    stereographic_projection(
        matrix_multiply(
            sphere(
                tau *
                token.get_macro("longitude") /
                36
                ,u
            )
            ,euler(
                0
                ,pi/2
                ,(2*pi/24) *
                token.get_macro("angle") /
                36
            )
        )
    ) [1] [2]
}
,z = 0*u
]
\appendcurve[
    u min = 0
    ,u max = {pi}
    ,u samples = 23
    ,transformation = {
        matrix_multiply(

```

```

        euler(
            0
            ,pi/2
            ,(2*pi/24) *
            token.get_macro("angle") /
            36
        )
        ,euler(pi/2,pi/3,pi/6)
    )
}
,x = {
    sphere(
        tau *
        token.get_macro("longitude") /
        36
        ,u
    ) [1] [1]
}
,y = {
    sphere(
        tau *
        token.get_macro("longitude") /
        36
        ,u
    ) [1] [2]
}
,z = {
    sphere(
        tau *
        token.get_macro("longitude") /
        36
        ,u
    ) [1] [3]
}
]
}
\foreach \latitude in {1,...,18}{
    \let\latitude\latitude
    \appendcurve[
        u min = 0
        ,u max = {2*pi}
        ,u samples = 4*45
        ,transformation = {euler(pi/2,pi/3,pi/6)}
        ,x = {
            stereographic_projection(

```

```

        matrix_multiply(
            sphere(
                u
                ,tau *
                token.get_macro("latitude") /
                36
            )
            ,euler(
                0
                ,pi/2
                ,(tau/24) *
                token.get_macro("angle") /
                36
            )
        )
    ) [1] [1]
}
,y = {
    stereographic_projection(
        matrix_multiply(
            sphere(
                u
                ,tau *
                token.get_macro("latitude") /
                36
            )
            ,euler(
                0
                ,pi/2
                ,(tau/24) *
                token.get_macro("angle") /
                36
            )
        )
    ) [1] [2]
}
,z = 0*u
]
\appendcurve[
    u min = 0
    ,u max = {2*pi}
    ,u samples = 45
    ,transformation = {
        matrix_multiply(
            euler(

```

```

0
,pi/2
,(2*pi/24) *
token.get_macro("angle") /
36
)
,euler(pi/2,pi/3,pi/6)
)
}
,x = {
sphere(
u
,tau*token.get_macro("latitude")/36
)[1][1]
}
,y = {
sphere(
u
,tau*token.get_macro("latitude")/36
)[1][2]
}
,z = {
sphere(
u
,tau*token.get_macro("latitude")/36
)[1][3]
}
]
}
\rendersegments
\end{tikzpicture}

```

pline is meant to be compatible with TikZ, and they are meant to be used together—at least for now. Maybe one day pline will have enough capabilities to do things like for loops and conditionals—at home. The code for Figure 3.1.3, which illustrates the use of a conditional, is given.

```

\begin{tikzpicture}
\clip
(-\textwidth/2,-5)
rectangle
(\textwidth/2,5)
;
\foreach[count = \c from 1] \angle in {20,40,...,80} {
\pgfmathsetmacro{\angle}{\angle*pi/180}
\let\angle\angle
\appendsurface[

```

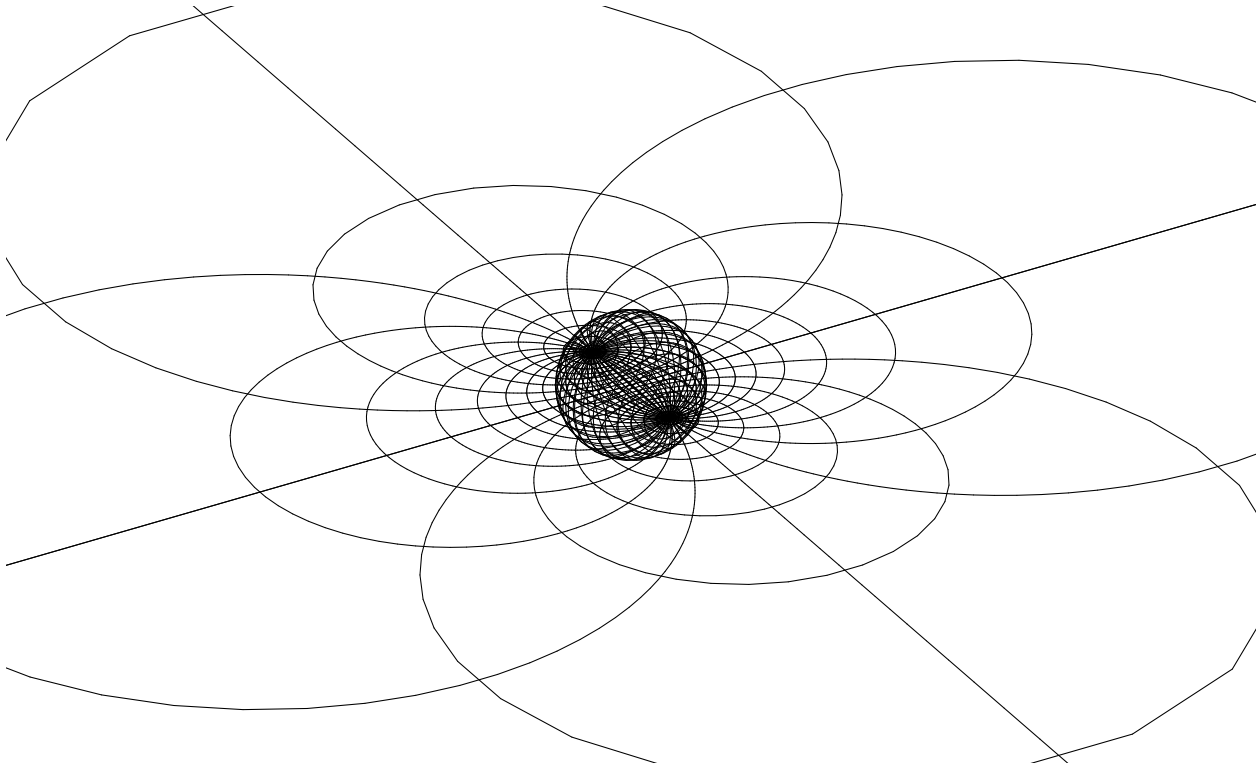


Figure 3.1.2:
Automatic division by zero handling.

```

u min = pi/6
,u max = 0.75*tau
,u samples = 36
,v min = 0
,v max = tau
,v samples = 36
,transformation = {euler(pi/2,pi/3,pi/6+pi)}
,draw options = {draw = black}
,fill options = {
    \ifnum\c=1
        fill = red
    \fi
    \ifnum\c=2
        fill = yellow
    \fi
    \ifnum\c=3
        fill = green
    \fi
    \ifnum\c=4
        fill = blue
    \fi
}

```

```

        ,fill opacity = 0.6
    }
    ,x = {
        (1 / cos(token.get_macro("angle"))) *
        cos(u) +
        sqrt(
            (
                1 /
                cos(token.get_macro("angle"))
            )^2 - cos(token.get_macro("angle"))
        ) *
        sphere(u,v)[1][1]
    }
    ,y = {
        (1 / cos(token.get_macro("angle"))) *
        sin(u) +
        sqrt(
            (
                1 /
                cos(token.get_macro("angle"))
            )^2 - cos(token.get_macro("angle"))
        ) *
        sphere(u,v)[1][2]
    }
    ,z = {
        sqrt(
            (
                1 /
                cos(token.get_macro("angle"))
            )^2 - cos(token.get_macro("angle"))
        ) *
        sphere(u,v)[1][3]
    }
]
}
\rendersegments
\end{tikzpicture}

```

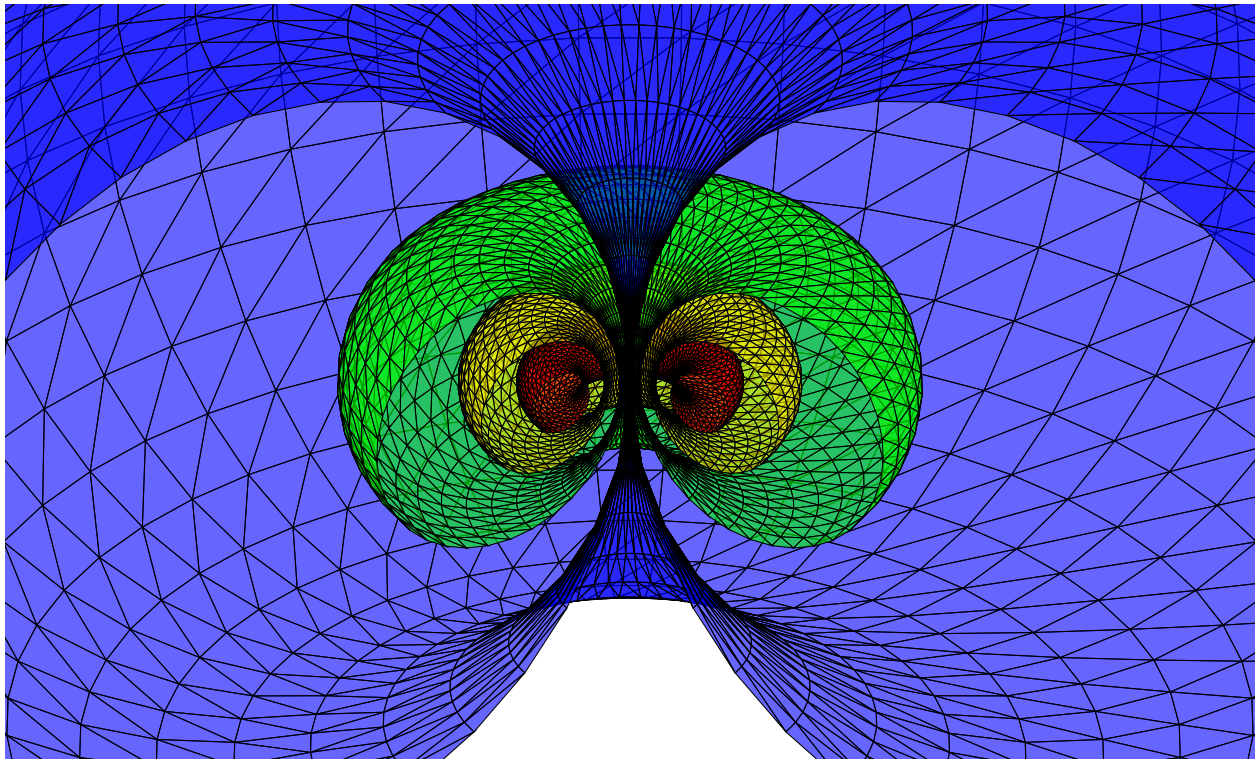


Figure 3.1.3:
Stereographically nested tori.

Chapter 4

Clipped Subspaces

A big focus for the future of this package is triangle-triangle clipping, which generalizes to polygon clipping. This dream is twofold; firstly it will enable parametric surfaces to intersect, and secondly it will enable intersecting plane diagrams which haven't been achieved in full generality in tikz yet.

4.1 Clipping individual planes.

This is where the technology is at currently. We can clip a single plane by a rectangular prism, and draw them. Eventually, when the polygon clipping gets going, this will evolve into a tool for graphing intersecting planes without using approximations. The code for Figure 4.1.1 is given.

```
\begin{tikzpicture}
  \appendprism[%
    a = 1
    ,b = 2
    ,c = 3
    ,d = 4
    ,x min = -1
    ,x max = 1
    ,y min = -1
    ,y max = 1
    ,z min = -1
    ,z max = 1
    ,transformation = {euler(pi/2,pi/3,pi/3)}
  ]
  \appendplane[%
    a = 1
    ,b = 2
    ,c = 3
    ,d = 1
    ,x min = -1
    ,x max = 1
    ,y min = -1
```

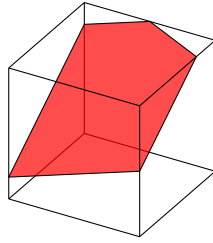



Figure 4.1.1:
A plane which has been clipped by a rectangular prism.

```

,y max = 1
,z min = -1
,z max = 1
,transformation = {euler(pi/2,pi/3,pi/3)}
]
\rendersegments
\end{tikzpicture}

```